

GENERIC MARKET INTELLIGENCE PLATFORM

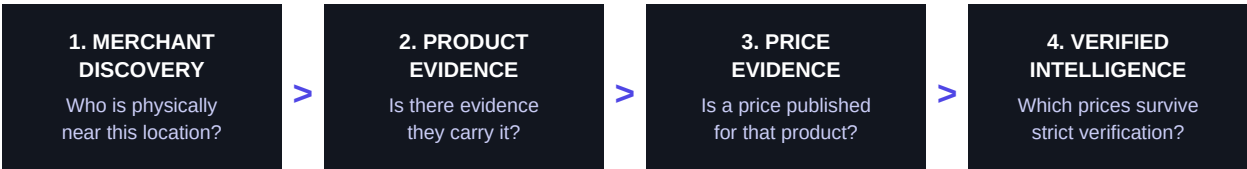
MarketRadar

Understand how a product is priced and positioned in a local market — and know exactly how much every number can be trusted.

Most price tools answer one question: **where is this cheapest?** MarketRadar answers a harder and more useful one: **what is actually happening in this market around this product and this location, and what is the evidence behind each claim?**

The design principle everything follows from

A merchant being discovered does not mean the product is available. The product being available does not mean a price exists. A price existing does not mean it is verified. MarketRadar keeps these four things separate and never lets one silently stand in for another. It prefers fewer trustworthy results over many incorrect ones.



Layer	Stack
Backend	Python, FastAPI, SQLAlchemy, SQLite
Frontend	React, TypeScript, Vite, Material UI
Market data	SerpApi (Google Maps, Shopping, organic search)
Verification	Custom product-identity and merchant-identity engines
Test suite	70 automated tests, no network or API credits required

Document purpose: an overview of what MarketRadar does, why it is built this way, and how to run it. Suitable for sharing.

1. The problem this solves

A business deciding how to price a product needs to know what the market around it looks like. The obvious approach is to search the web, collect the prices that come back, and compare. That approach quietly produces wrong answers, and the reasons are structural rather than accidental.

Four failures that look like data

These are not hypothetical. Each one was observed in real search-provider responses while building this system.

What goes wrong	Why it is dangerous
The price belongs to a different product. A search result page lists twenty products and one price. Scraping the price off that page attaches it to whichever product you were looking for.	You get a confident, specific, wrong number. There is nothing in the output to suggest it is wrong.
The result is not from that merchant at all. A site-restricted search that finds nothing falls back to general web results. Those results then get attributed to the merchant you were searching.	A competitor appears to stock a product based on a page that has no connection to them.
The size or variant differs. A 50 ml miniature, a 1 L bottle and a 4-pack are not the same product as a 750 ml bottle. Neither is a Pro Max the same as a Pro.	Averages built from mixed variants describe a market that does not exist.
A failed request reports as zero. When a provider call errors and the error is swallowed, the interface shows "0 results found".	Absence of evidence is displayed as evidence of absence — the most misleading failure of the four.

Why this matters more than raw coverage

A pricing decision made on a fabricated number is worse than a pricing decision postponed for lack of data. A system that says "I could not verify this" is more useful than one that guesses confidently. That judgement shapes every rule below.

2. How MarketRadar works

A Price Intelligence search takes three inputs — a product, a location and a radius — and runs them through four stages that are never collapsed into each other.

Stage 1 — Merchant discovery

The location is geocoded into an explicit search centre, and merchants are discovered around that point. Distance is computed from the resolved centre to each merchant's own coordinates, and the radius filters the set. The search centre is never guessed: if a location cannot be resolved, the request fails with a clear error rather than falling back to some default point, because a wrong centre silently corrupts discovery, distance, filtering and caching all at once. This stage establishes only that a merchant exists nearby.

Stage 2 — Product evidence

For each discovered merchant, the system looks for evidence that the requested product exists in their catalogue. Every result is checked to confirm it is actually served by that merchant's own domain before it counts as evidence from them. Product identity is established from the page title and URL path only — never from the result snippet, because a snippet describes a page rather than a product. The outcome is one of three states: **found**, **not found**, or **unknown**. "Unknown" is a real answer, not a failure.

Stage 3 — Price evidence

Price is assessed separately from availability, and carries its own three states: **verified**, **unavailable**, or **unknown**. A price is only verified when it is unambiguously tied to the requested product. Prices are rejected when they come from a category or brand listing page, when the source shows a range or a "starting at" figure, when several unrelated prices appear together, when the requested size cannot be confirmed, or when the merchant identity is uncertain. A merchant that clearly stocks a product but publishes no price is recorded as *product found, price unavailable* — never as "not sold".

Stage 4 — Verified intelligence

Only verified observations enter the statistics. Lowest, average and highest prices are computed from that set alone. Everything excluded is still shown, with the specific reason it was excluded, so the gaps in the market picture are visible rather than hidden.

Evidence scope: catalogue is not inventory

A chain merchant has one website and many physical stores. Finding a product in that chain's online catalogue proves the chain lists it. It does not prove the branch four miles away has it on the shelf. MarketRadar records this distinction explicitly on every observation, so a catalogue listing is never presented as confirmed local stock.

3. The rules that make results trustworthy

These are enforced in code and covered by the test suite.

Rule	What it means in practice
Discovery broad, verification strict	Searching may cast a wide net across naming variants. Confirming a match may not. Recall problems are never solved by loosening verification.
Provenance is checked	A result counts as evidence from a merchant only if it is genuinely served by that merchant's domain.
Identity from title and URL, never snippet	A page blurb proves the page is relevant. It does not establish which product the page is about.
Meaningful attributes are preserved	Harmless differences normalise together ("750 ML" equals "750ml", "15 Yr" equals "15 Year"). Identity-changing ones never do: Pro is not Pro Max, 256 GB is not 128 GB, 750 ml is not 1 L, a 4-pack is not a single unit.
Uncertain merchant identity cannot verify	If evidence cannot be tied to one specific nearby store — for instance when several branches of a chain are in range — no verified store-level price is produced.
A failed stage is never a zero	Every response reports the status of each stage. A provider that errored says so explicitly instead of quietly reporting no results.

Rule	What it means in practice
Statistics admit only verified data	Ambiguous, missing, wrong-size, wrong-variant and uncertain-merchant observations are excluded from every aggregate.
No mock data in production paths	Fixtures live in tests. Application code contains no fabricated merchants, prices, distances or availability.
Industry-agnostic by construction	No merchant, brand, product, category or location is special-cased. The same code paths serve phones in Hyderabad, trainers in New York and groceries in London.

4. What this looks like in practice

A worked example from a real search — a spirits product across a 25-mile radius — run against genuine cached provider responses. It shows how a raw result count collapses into a small number of claims that survive scrutiny.

Funnel step	Count	What the number means
Nearby merchants discovered	20	Physical merchants inside the radius, measured from the resolved search centre.
Raw indexed results returned	37	Everything the search provider returned across those merchants.
Rejected as off-domain	10	Results that were not served by the merchant domain at all — video, forum and film pages returned as fallbacks.
Genuine merchant results	27	Confirmed as served by the merchant's own site.
Excluded as listing pages	8	Category and brand pages whose prices cannot be tied to one product.
Verified local prices	1	Survived every check, traced to a single product page.

One verified price out of thirty-seven results is the point, not a shortfall

Before these rules were enforced, the same data produced several "verified" prices lifted from brand listing pages — numbers that belonged to no particular product. The honest output is one price that can be traced to a real product page, plus a clear account of why the other thirty-six did not qualify.

Every merchant that did not yield a verified price is still reported individually, with its own product status, price status and a specific reason — price not product-specific, size unverified, merchant identity uncertain, price unavailable, and so on. The result is a map of the local market that includes its own blind spots.

5. Using MarketRadar

Requirements

Python 3.9 or later, Node.js, and a SerpApi API key.

Backend

```
cd backend
python -m venv venv
venv\Scripts\activate          # Windows
pip install -r requirements.txt -r requirements-dev.txt
copy .env.example .env         # then set SERPAPI_API_KEY
uvicorn app.main:app --reload --port 8001
```

Health check: <http://127.0.0.1:8001/api/health> • Interactive API docs: <http://127.0.0.1:8001/docs>

Frontend

```
cd frontend
npm install
npm run dev
```

Served at <http://localhost:5173>. The backend must be running on port 8001.

Tests

```
cd backend
pytest
```

70 tests, no network calls and no API credits consumed. Collection is restricted to the tests directory, and the suite is pinned to an isolated test database that can never be the application database.

A Price Intelligence request

```
GET /api/prices/local
    ?product=Apple iPhone 15 Pro 256GB
    &area=Hyderabad, Telangana, India
    &radius=25
```

The response carries the resolved search centre, the status of every stage, each nearby merchant with its independent product and price status, the verified observations with their sources, and the aggregate statistics. An area that cannot be geocoded returns an explicit error rather than a result computed from a guessed location.

Module status

Stated plainly, because documenting planned work as finished is its own kind of fabricated data.

Module	Status	Description
Dashboard	Live	Entry point and navigation across modules.
Price Intelligence	Live	The full four-stage evidence funnel described in this document.
Market Research	Live	Search visibility: organic results, local results and the domains appearing in a market.
Competitors	Scaffolded	Turning search and merchant evidence into ranked competitor candidates.
Products	Scaffolded	Product-level view: merchants carrying it, observed prices, price history.
Trends	Scaffolded	Change over time. Requires repeated snapshots; a single search is not a trend.

Module	Status	Description
Market Pulse	Scaffolded	Combined current-market signal across the other modules.
AI Analyst	Scaffolded	Analysis layer over collected evidence, separating fact from inference.

Scaffolded modules are routed, navigable and backed by registered endpoints. They render honest empty states rather than placeholder metrics.

6. Who this helps, and how

Situation	What MarketRadar gives you
Setting a price for a new product	The verified local price range, the number of nearby merchants that actually carry it, and an explicit account of where evidence is missing.
Watching a competitor	Which nearby merchants show evidence of a product, which publish prices, and which need manual investigation.
Entering a new area	Merchant density within a chosen radius of a specific point, and how visible the category is in search there.
Auditing your own market position	An optional reference retailer is compared against competitor observations while being kept out of the competitor statistics.
Deciding where to look harder	Every unverified merchant carries a specific reason, which turns "no data" into a prioritised list of things worth checking manually.

Because nothing in the production logic is tied to an industry, the same pipeline applies to electronics, groceries, fashion, appliances, beverages, sporting goods or automotive parts. Product attributes are matched dynamically from whatever exists — brand, model, capacity, size, variant, pack count — rather than from a fixed schema.

What it deliberately does not do

Being explicit about limits is part of being trustworthy.

- It does not claim complete market truth. It reports observed public signals and labels the confidence behind each.
- It does not confirm physical shelf stock. Catalogue evidence is labelled as catalogue evidence.
- It does not infer trends from a single search. Trend claims require repeated observations over time.
- It does not bypass access controls, rate limits or bot protection. Evidence comes from public search results.
- It does not fill gaps with estimates. Where evidence is insufficient, it returns "unknown".

7. Summary for sharing

A short version of the above, ready to post or send. Adapt freely — every fact in it is accurate as written.

*I have been building **MarketRadar**, a market intelligence platform that answers "what is happening in this market around this product and location" rather than just "where is this cheapest".*

The hard part turned out not to be collecting data. It was refusing to trust most of it.

A few things I found while building the verification layer:

- A site-restricted search that finds nothing silently falls back to general web results. Ten of thirty-seven "merchant results" in one real query were video, forum and film pages.*
- Prices scraped from brand and category listing pages look perfectly valid and belong to no particular product.*
- A failed API call that gets swallowed shows up in the UI as "0 results" — absence of evidence rendered as evidence of absence.*

So the architecture separates four things that are usually collapsed together: merchant discovered, product available, price published, price verified. Each has its own state, and one never stands in for another.

On that real query, thirty-seven raw results reduced to one verified price. That is the feature. The other thirty-six are reported with the specific reason they did not qualify, so the gaps are visible instead of hidden.

Fewer trustworthy results beat many incorrect ones — especially when someone is about to make a pricing decision from them.

Python / FastAPI / React / TypeScript. Industry-agnostic by design: the same pipeline handles phones, groceries or spirits without special-casing any of them.

MarketRadar — merchant discovery, product evidence, price evidence, verified market intelligence. Every result traceable to the evidence behind it.